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Proteomic analysis of rat laryngeal muscle following denervation

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Laryngeal muscle atrophy induced by nerve injury is a major factor contributing to the disabling symptoms associated with laryngeal paralysis. Alterations of global proteins in rat laryngeal muscle following denervation were, therefore, studied using proteomic techniques. Twenty-eight adult Sprague-Dawley rats were divided into normal control and denervated groups. The thyroarytenoid (TA) muscle was excised 60 days after right recurrent laryngeal nerve was resected. Protein separation and identification were preformed using 2-DE and MALDI-MS with database search. Forty-four proteins were found to have significant alteration in expression level after denervation. The majority of these proteins (57%), most of them associated with energy metabolism, cellular proliferation and differentiation, signal transduction and stress reaction, were decreased levels of expression in denervated TA muscle. The remaining 43% of the proteins, most of them involved with protein degradation, immunoreactivity, injury repair, contraction, and microtubular formation, were found to have increased levels of expression. The protein modification sites by phosphorylation were detected in 22% of the identified proteins that presented multiple-spot patterns on 2-D gel. Significant changes in protein expression in denervated laryngeal muscle may provide potential therapeutic strategies for the treatment of laryngeal paralysis.

Keywords:
Denervation / Laryngeal muscle / Muscle atrophy

1 Introduction

Recurrent laryngeal nerve injury results in vocal fold paralysis or paresis with subsequent voice change, difficulty in swallowing, and possibly difficulty in breathing. Laryngeal muscle atrophy occurring as a result of denervation worsens these symptoms. It is generally believed that the primary mechanism of muscle atrophy is a process of selective protein loss by both decreasing protein synthesis and increasing protein degradation; however, selective increases in protein expression have been observed at both transcription and translation levels during the processes of muscle atrophy [1, 2]. To better understand the complex processes of protein synthesis and degradation during muscle atrophy, analysis of global protein expression will be required.

Studies focused on muscular contractile protein demonstrate that laryngeal muscles have the unique character of myosin heavy chain (MyHC) isoform expression under both normal and denervated conditions in comparison to limb muscles. The limb muscle normally expresses fast IIA, IIX, IIB, and slow type I MyHC. During the process of limb muscle denervation, the level of atrophy observed in fast

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Abbreviations: CSP, common salivary protein; **EOM**, extraocular muscle; **GAPDH**, glyceraldehyde 3-phosphate dehydrogenase; **HMG-CoA**, 3-Hydroxy-3-methylglutaryl-coenzyme A; **JIP**, C-jun-amino-terminal kinase interaction protein; **MHC**, major histocompatibility complex; **MLC**, myosin light chain; **MyHC**, myosin heavy chain; **PNP**, purine nucleoside phosphorylase; **SHF**, heat shock transcription factor; **TA**, thyroarytenoid

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muscle fibers (IIA/IIX and IIB) appears to be greater than that observed in slow muscle fibers (type I), with a predominance of slow fibers in the denervated limb muscle [3, 4]. To serve the highly specialized and complex function of coordinating vocal fold motion for airway protection, respiration phonation, laryngeal muscle has evolved into faster contractility. For example, the thyroarytenoid muscle (TA), the primary muscular component of the vocal fold, possesses the capacity to express specific extraocular muscle (EOM) MyHC. EOM MyHC is a faster contractile protein compared to fast IIA, IIX, and IIB MyHC in limb muscles fibers and not expressed in limb muscle [5, 6]. Fast IIA and IIX MyHC, a common expression MyHC in limb muscle, is not generally present in TA muscle [6, 7]. In denervated laryngeal muscle, EOM MyHC is decreased and fast IIX/IIA MyHC appears. Its appearance in denervated TA muscle is in contrast to its loss in denervated limb muscle. Although unique changes of major histocompatibility complex (MHC) components have been described in denervated laryngeal muscle, little is known about alterations occurring in the general population of muscle proteins.

The current study utilizes 2-DE and MS to explore the complex network of protein expression in laryngeal muscle atrophy induced by denervation. These results provide information about the expression complementary to MyHC alteration during progression of TA muscle atrophy. Moreover, these results will provide researchers and clinicians with a better understanding of the mechanisms affecting fast type muscle atrophy induced by denervation.

2 Materials and methods

2.1 Denervation of rat laryngeal muscle

Twenty-eight 220–250 g male Sprague-Dawley rats were divided into control and denervated groups. The rats were anesthetized using intramuscular injection of ketamine hydrochloride (35 mg/kg) and xylazine hydrochloride (5 mg/kg). Using an operating microscope, the right recurrent laryngeal nerve was isolated through a midline incision. For the control group, the incision was sutured after the nerve was identified. For the denervation group, an approximately 2 mm segment of the nerve was resected, and the proximal and distal nerve ends ligated to prevent reinnervation, and then the incisions were closed. After survival for 60 days, the rats were euthanized using a lethal dose of sodium pentobarbital and the specimens were harvested immediately.

2.2 Histological section

In order to determine the degree of muscle fiber atrophy, the larynges of two rats from each group were excised and immediately frozen with OTC in isopentane cooled in dry ice. Fifteen micrometer thick cross sections were cut at −20°C and Hematoxylin-eosin (HE) staining was performed. Cross-sectional areas of 300 fibers were determined from three fields within each section using NIH-image 1.62 software (Scion Corporation).

2.3 2-DE

The right TA muscle was excised from ten rats in each group, and then was snap-frozen in liquid nitrogen. Due to small size of the specimens, TA muscles from two rats were homogenized together in sample buffer (7 M urea, 2 M thiourea, 4% ASB-14, 0.5% Triton X-100, 10 mM DTT, 0.5% carrier ampholyte pH 3–10 or 5–8), yielding five muscle samples in each group. The homogenates were centrifuged at 25 000 rpm for 60 min at 4°C. Total protein concentration of supernatant was determined using DC protein assay kit (BioRad). The sample buffer was added to adjust equal concentration of each sample. The supernatant was divided into aliquots and stored at −80°C.

IEF was performed using a PROTEAN IEF Cell electrophoresis system (BioRad) at 20°C. Seventeen-centimeter ReadyStrip IPG strips with a pH range of 3–10 and 5–8 (BIO-RAD) were used for the determination of optimal isoelectric range. The pH 5–8 strips were used for all experimental samples. The strips were passively rehydrated overnight with 100 μg total protein sample for silver staining gel and 200 μg for CBB staining gel. The linear ramping mode of IEF voltage was applied according to the following paradigm: 300 V for 60 min, 600 V for 60 min, and 1500 V overnight, followed by 10 000 V to achieve 60 kV/h at 20°C. After IEF, the strips were kept in −80°C.

For second dimension electrophoresis, the strips were first equilibrated at room temperature for 15 min in equilibration solution I (6 M urea, 2% SDS, 375 mM Tris-HCl pH 8.8, 20% glycerol, 30 mM DTT) and then transferred to equilibration solution II for 15 min (6 M urea, 2% SDS, 375 mM Tris-HCl pH 8.8, 20% glycerol, 135 mM iodoacetamide). The electrophoresis was performed at 12 mÅ/gel, 4°C, using a 12% separating gel and 4% stacking SDS page gel and was terminated when the bromophenol blue dye reached the bottom of the gel. The gels were stained with silver [8] or CBB G-250 (BioRad).

2.4 Protein spot analysis

PDQuest software (BioRad) was used for spot detection, quantification and statistical analyses. The matching errors were manually evaluated. The spot MW and p*I* were automatically calculated according to molecular weight marker and strip p*I*. The intensity of each spot was normalized by the total intensity of all the valid spots on the entire gel. The units of spot intensity used "Intensity*Area in Parts *per* Million" (INT*Area PPM). The silver stained gels and CBB stained gels were separately analyzed for comparison of normal and denervation rats.

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2.5 Protein spot identification

Protein spots with significant ($p < 0.05$) changes in density after denervation were cut from the gel in 1 × 1 mm samples. Four similar control samples were cut from nonstaining areas on each gel. Individual gel samples were placed in 1.5 mL microcentrifuge tubes, prerinsed with ACN, and washed twice for 10 min with 50% methanol. Sliver stained gel samples were placed in a 1:1 mixture of 30 mM K ferricyanide, and 100 mM Na thiosulfate until the gel pieces turned clear. Each gel sample was washed in a Milli-Q of water with three quick rinses, followed by rinses with 50 mM ammonium bicarbonate and ACN for 10 min each. Gel samples were resuspended in 10 mM DTT in 100 mM ammonium bicarbonate and incubated for 1 h at 56°C. After cooling to room temperature, the gel pieces were placed in 55 mM iodoacetamide in 100 mM ammonium bicarbonate for 45 min at room temperature in the dark. Samples were again washed twice for 10 min in 50 mM ammonium bicarbonate and ACN. Gel samples were completely dried by vacuum centrifugation and resuspended in trypsin solution (10 ng in 50 mM ammonium bicarbonate) on ice for 45–60 min. Excess trypsin solution around each sample was discarded and gels were then covered with 25 μL of 50 mM ammonium bicarbonate. Following overnight incubation at 37°C, the supernatant was collected in a second prerinsed microcentrifuge tube. The gel samples were extracted once in 25 mM ammonium bicarbonate for 20 min and three times in 50% ACN and 5% formic acid. The supernatants were combined and dried under vacuum centrifugation, and peptide extracts were stored at –20°C.

Peptide extracts were resuspended in 5 μL of 50% ACN and 0.3% TFA. One microliter of aliquots were mixed with equal volumes of CHCA dissolved in 50% ACN and 0.3% TFA, and applied to the target wells. MALDI-MS spectra were obtained on a Voyager DE-STR MALDI-TOF mass spectrometer (PE Biosystems Voyage system 4199) in the positive-ion reflector mode with delayed ion extraction. An accelerating voltage of 20 kV and 65% grid voltage were employed, and acquisition mass range was 800–3000 Da with low mass gate set at 500 Da. An external peptide mixture was used to calibrate the instrument. Monoisotopic peptide masses were assigned to search for protein using the MS-Fit module of the ProteinProspector program supplied by PerSeptive Biosystems. All proteins with positive identification matched a minimum of four peptides within 50 ppm, allowing one missed cleavage.

2.6 LC MS/MS and RT-PCR

Four samples used in MALDL-MS were performed by LC MS/MS. Nanoflow HPLC is Famos autosampler, Switchos, and Ultimate micropump. Column C18 from LC Packings (75 μm × 10 cm) and gradient from 2 to 100% methanol over 20 min were employed. Flow rate was setup 200 nL/min. MS is API QSTAR Pulsar from PE Sciex/API. NanoSpray voltage was 1500 V. TOF-MS survey scan was at 350–1200 amu for 0.5 s, and product MS was 65–1200 amu for 1.2 s. Collision energy was applied at 25/30/35 for each parent ion selected. MASCOT (Matrix Science) software was used in date analysis.

RT-PCR was used to investigate mRNA expression of common salivary protein-1 (CSP-1), which has been confirmed by both MALDI-MS spectra and LC MS/MS. Total RNA was isolated from two normal and two denervated rat TA muscle by using Trizol Reagent (Invitrogen). The primer design, based on Rattus norvegicus CSP-1 (accession No. NM-133622), includes the primer sequences GCTCCATGCTGCCCTTACTGATT (forward) and CTCTGT ACATGTACCGTTTACAT (reverse). The expression of glyceraldehyde 3-phosphate dehydrogenase (GAPDH) mRNA was used for normalization control. GAPDH primers are CAACTACATGGT (forward) and GCCAGTGGACTCCAC-GAC (reverse). Open read frame of CSP-1 was amplified using SuperScript™ one-step RT-PCR for long templates kit (GibcoBRL).

2.7 Statistical analysis

All statistical comparisons were performed using the unpaired two-tailed Student's *t*-test. The statistical significance of $p < 0.05$ was accepted.

3 Results

3.1 Changes in muscle fiber cross-sectional area

The denervated group demonstrated characteristic morphologic changes associated with muscle atrophy. Muscle fibers were loosely packed with increased number of nuclei. Fiber size was variable, but the fibers in the denervated group were consistently smaller than in controls, with cross-sectional areas approximately 46% less than those in the control group ($p < 0.05$, Fig. 1).

3.2 IPG strip selection and optimization

2-DE was performed in broad and narrow p*I* ranges to determine the optimal isoelectric range where the maximum number of protein spots can be detected on a single gel.

Five normal TA muscle samples were studied using pH 3–10 and 5–8 range strips. Spots with density great than 150 INT*Area PPM were counted. Using pH 3–10 IPG strip, 444 ± 54 protein spots were displayed on single silver stained gel with 313 ± 38 (70%) of the spots distributed between the 5 and 8 p*I* range. Using a pH 5–8 IPG strip, 459 ± 63 protein spots were detected, representing a 47% increase compared to the number of spots in the same pH region on the pH 3–10 IPG strip (Fig. 2). Because of the higher efficiency, pH 5–8 IPG strips were used for protein separation.

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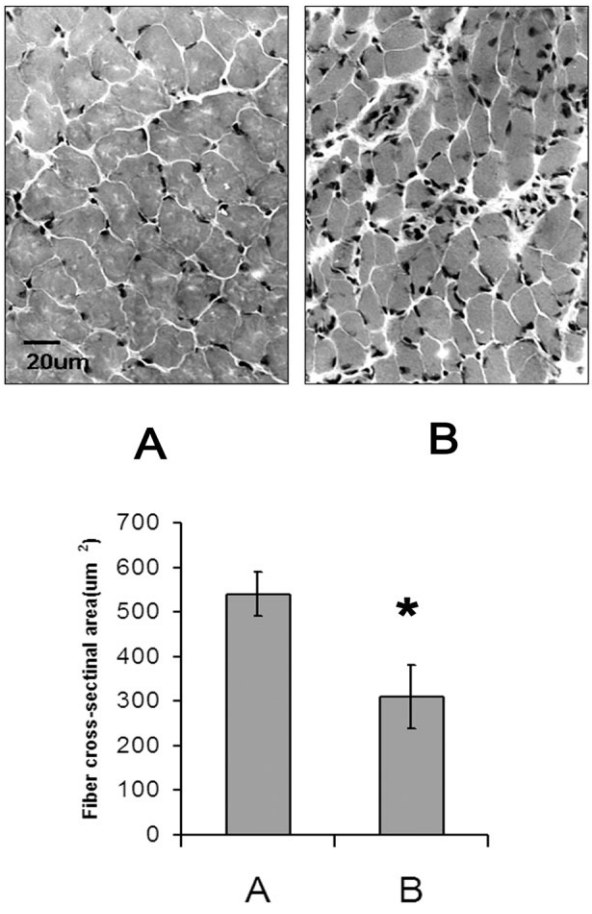


Figure 1. Histological comparison of cross-sectional area (40× HE) between control rat TA muscle fibers (A) and denervated muscle fibers (B). Bar figure represents mean of cross-sectional areas. $*p < 0.05$.

3.3 Identification of protein spots that changed significantly

All samples were assessed using pH 5–8 IPG strips. Each sample was run two gels, one stained with sliver and another stained with CBB. A total of 537 spots were matched on silver stained gels and a total of 388 spots were matched on CBB stained gels. Forty-four protein spots changed densities (19 spots increased and 25 spots decreased) in the denervated group rats ($p < 0.05$). These protein name, p*I*, MW, function and change in density are provided in Table 1. These spot locations on 2-D gel are demonstrated in Fig. 3. Examples of the spots with increased and decreased densities on 2-D gel are shown in Fig. 4.

3.4 Confirmation of protein spots

Four protein spots were selected for LC MS/MS for confirmation of MALDI-TOF MS results. The LC MS/MS finding yielded the same identification as MALDI-TOF (Table 2). CSP-1, one of the four, was selected for RT-PCR. Expression of CSP-1 mRNA was increased to 82% in denervated TA muscle compared to control. The alteration in transcription level detected by RT-PCR is consistent with the expression of CSP-1 protein detected by 2-DE (Fig. 5).

3.5 Alteration of multiple-spot patterns

Seven proteins exhibited unique multiple-spot patterns with different p*I* and MW. Frequent differences in spot intensity between the control and denervated TA muscles were observed. Twenty-nine phosphorylated amino acid sites were identified from 10 spots associated with 4 proteins (Table 3). An example of 2-D gel image demonstrating multiple spots and the corresponding mass spectrum of phosphorylated peptides are shown in Fig. 6.

4 Discussion

4.1 The process of laryngeal muscle atrophy induced by denervation

Limb skeletal muscle atrophy begins in the first few days following denervation. From 2 wk to 4 months, denervated muscle loses more than 40% of their mass [9–11]. Early nuclear proliferation occurs within 8 wk [12]. In the current study, fiber cross-sectional area in the rat TA muscle was decreased by 46% while the nuclear number was increased after 2 month denervation. These results are consistent with the studies of limb muscle atrophy described above and with previous studies of laryngeal muscle, where MyHC was used to describe atrophy following denervation [13].

4.2 Proteins associated with denervation-induced TA muscle atrophy

This study identified 44 protein spots including phosphorylated isoforms associated with denervation-induced atrophy. Fifty-seven percent of the spots exhibited decreased expression levels and the other 43% showed increased expression levels. The proteins identified in this study are involved in a variety of functions: energy metabolism (29%), cell proliferation and differentiation (11%), regulation of muscle contraction (13%), pathway of degradation of cellular proteins (2%), stress reaction (2%), immunoreactivity and injury repair (4%), signal transduction and membrane (11%), microtubule nucleation (9%), and other miscellaneous functions (18%).

4.2.1 Enzymes associated with energy metabolism

Thirteen of the identified protein spots are associated with energy metabolism. Similar percent distribution of this category's proteins was reported in normal and atrophied rat limb muscle as well as human laryngeal muscle [14–17]. Expression level decreased in nine spots (77%) and increased in two (15%).

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Table 1. Proteins identified in rat TA muscle with altered expression after denervation compared to control

Spot no.[a)]	Protein name and NCBI accession no.	Theoretical MW (kDa/p*I*)	Experimental MW (kDa/p*I*)	Peptide cover (%)	Spot density, (INT*Area PPM[b)])		Changes, (%)	*F*[c)]
					Denervation	Control		
B4b	Lactate dehydrogenase (NP_623521)	40/6.3	43/6.7	16	387.9 ± 59	491.01 ± 88	−21	1
B4c	β-enolase (P15429)	46/7.74	42/6.7	10	277.54 ± 43	923.33 ± 57	−70	1
B4d	β-enolase (P15429)	40/6.45	39/6.7	23	1755.09 ± 176	2166.67 ± 174	−19	1
B6	Common salivary protein 1 (19424346)	17/7.7	18/6.4	50	1078.4 ± 442	73.06 ± 35	1376	9
B6a	α-enolase (17367186)	47/6.0	47/6.4	18	147.71 ± 31	2954.20 ± 245	−95	1
B6b	α-enolase (17367186)	47/6.2	47/6.6	23	1945.49 ± 132	3413.14 ± 284	−43	1
B7	Common salivary protein 1 (19424346)	17/7.7	19/6.8	56	1211.33 ± 636	308.23 ± 61	293	9
B7a	Isocitrate dehydrogenase subunit Alpha (2851463)	37/6.0	42/6.4	30	2053.89 ± 212	2892.81 ± 198	−29	1
B7b	β-enolase (P15429)	46/7.58	42/6.5	22	265.23 ± 48	3312.53 ± 267	−92	1
B8a	DNA polymerase III (16126066)	35/6.41	30/6.6	19	288.05 ± 27	3600.62 ± 252	−92	2
B9a	Acetyl-Coenzyme A dehydrogenase (NP_031407.2)	47/8.34	43/7	18	1434.42 ± 115	1815.72 ± 143	−21	1
B9b	Creatine kinase (NP_036662.1)	43/6.6	43/7.5	39	4166.49 ± 364*	4340.09 ± 569	− 4	1
B9c	Creatine kinase (NP_036662.1)	43/6.6	43/7.6	39	24145.80 ± 1253	13414.33 ± 562	80	1
B18a	Vacuolar ATP synthase subunit H (P41807)	54/5.9	56/6.2	20	914.22 ± 48	397.48 ± 16	130	1
B18b	Low Km (high- affinity) cAMP phosphodiesterase (6324936)	61/6.1	53/6.3	11	497.79 ± 37	123.21 ± 42	304	7
S	Myosin light chain 1 (127129)	20/5	22/5.5	25	70213.29 ± 2112	49445.98 ± 987	42	3
S1	Membrane protein 7 (16082292)	30/5.7	25/5.5	20	1778.12 ± 759	4679.26 ± 299	−62	7
S2	NADH-ubiquinon oxidoreductase (20884969)	24/5.9	24/5.6	20	2710.50 ± 275	6610.98 ± 513	−59	1
S3	Myosin light chain 1 (127131)	20/5	20/5.6	20	1144.72 ± 742	14309 ± 582	−92	3
S4	Myosin light chain 3 (6981240)	22/5	25/5.5	56	1711.98 ± 479	3356.82 ± 253	−49	3
S5	3-hydroxy-3-methylglutaryl-coenzyme A reductase (40538852)	38/8.51	34/6.8	32	481.47 ± 61	8024.5 ± 582	−94	2
S5a	Pir protein (14970560)	30/5.5	32/6.1	29	1386.94 ± 231	402.01 ± 21	245	9
S5b	5,10 methylenetetrahydrofolate reductase (15794085)	33/6	30/6.1	30	3169.62 ± 196	6096.43 ± 427	−48	9
S6	Unknown protein (19347803)	28/4.64	28/6.7	28	908.75 ± 119	3495.19 ± 427	−74	9
S9	MHC class II beta (1002962)	17/6.6	13/7.2	38	8452 ± 430	470.34 ± 232	1697	6
S10	OspD1 (18462546)	25/6.5	28/6.8	24	1735.91 ± 132	2942.22 ± 168	−41	9
S11	Hypothetical protein (19745986)	34/7.1	34/7.3	23	578.18 ± 41	2223.72 ± 153	−74	9

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Table 1. Continued

Spot no.[a)]	Protein name and NCBI accession no.	Theoretical MW (kDa/p*I*)	Experimental MW (kDa/p*I*)	Peptide cover (%)	Spot density, (INT*Area PPM[b)])		Changes, (%)	*F*[c)]
					Denervation	Control		
S12	Tumor-associated calcium signal transducer 1 precursor (120749)	34/7.4	33/7.3	14	715.75 ± 57	2752.87 ± 178	−74	7
S13	Unknown protein (9836834)	44/6.32	34/6.9	18	1144.48 ± 89	454.16 ± 38	152	9
S14	RAB13 (Member RAS oncogene family)(NP_006859)	21/6.6	22/6	30	616.89 ± 49	3628.76 ± 255	−83	7
S14b	Purine-nucleoside phosphorylase (2143548)	32/6.1	32/7	21	1735.87 ± 576	304.53 ± 59	470	6
S16	c-jun-amino-terminal kinase interacting protein 1 (20558919)	28/6.6	22/6.2	19	1965.64 ± 117	922.84 ± 71	113	7
S17a	Triosephosphate isomerase 1 (12621074)	27/6.5	28/7.2	43	3293.22 ± 283	6333.11 ± 433	−48	1
S17b	Ras-like GTP binding protein (477814)	24/6.2	26/7.2	28	3624.65 ± 289	5034.23 ± 476	−28	2
S17d	Replication-associated protein (19033331)	26/7.8	24/7.2	32	2426.71 ± 132	8089.90 ± 542	−70	2
S19a	γ-tubulin complex component 2 (Q9BSJ2)	102/6.3	100/6.2	10	3425.46 ± 235	827.41 ± 75	314	8
S19b	γ-tubulin complex component 2 (Q9BSJ2)	102/6.3	100/6.3	13	3007.15 ± 241	586.18 ± 73	413	8
S19c	γ-tubulin complex component 2 (5729840)	102/6.3	100/6.4	14	1797.85 ± 346	531.91 ± 48	238	8
S19d	γ-tubulin complex component 2 (Q9BSJ2)	102/6.3	100/6.5	14	2339.97 ± 274	359.99 ± 42	550	8
S20a	Heat shock factor protein 24 (123684)	33/5.8	33/6.2	20	1190.93 ± 87	2646.51 ± 167	−55	5
S20b	Proteasome 28 subunit (6755212)	28/5.7	31/6.1	29	1463.84 ± 131	1030.87 ± 89	42	4
S20c	DNA-directed RNA polymerase alpha subunit (20808632)	35/5.5	30/6.1	21	685.77 ± 52	2078.09 ± 156	−67	2
S21a	Parvalbumin (Calcium binding protein) (11968064)	12/5	12/5	42	38541.56 ± 1241	11335.75 ± 764	240	3
S21b	Parvalbumin (Calcium binding protein) (11968064)	12/5	12/5.2	34	9466.71 ± 623	2809.11 ± 209	237	3
S21d	Myosin light chain 2 (7949078)	18/4.8	18/5	40	50734.83 ± 1052	19896.01 ± 772	155	3

a) Spot labeled as B were identified from CBB stained gels and spots labeled as S were identified from silver stained gels. Silver stained gels and CBB stained gels were separately preformed statistical analyses by using PDQuest software.

b) INT*Area PPM: Intensity*Area in Parts *per* Million was spot density unite. All spots in denervated groups have a significant alteration in densities ($p < 0.05$), except for B9b labeled *.

c) Protein function groups: 1. Energy metabolism proteins; 2. Cell proliferation and differentiation proteins; 3. Protein for regulation of muscle contraction; 4. Proteins in pathway of degradation of cell; 5. Stress protein; 6. Immunoreactivity and injury repair proteins; 7. Signal transduction and membrane proteins; 8. Mircotubule nucleation proteins; 9. Other and unknown proteins.

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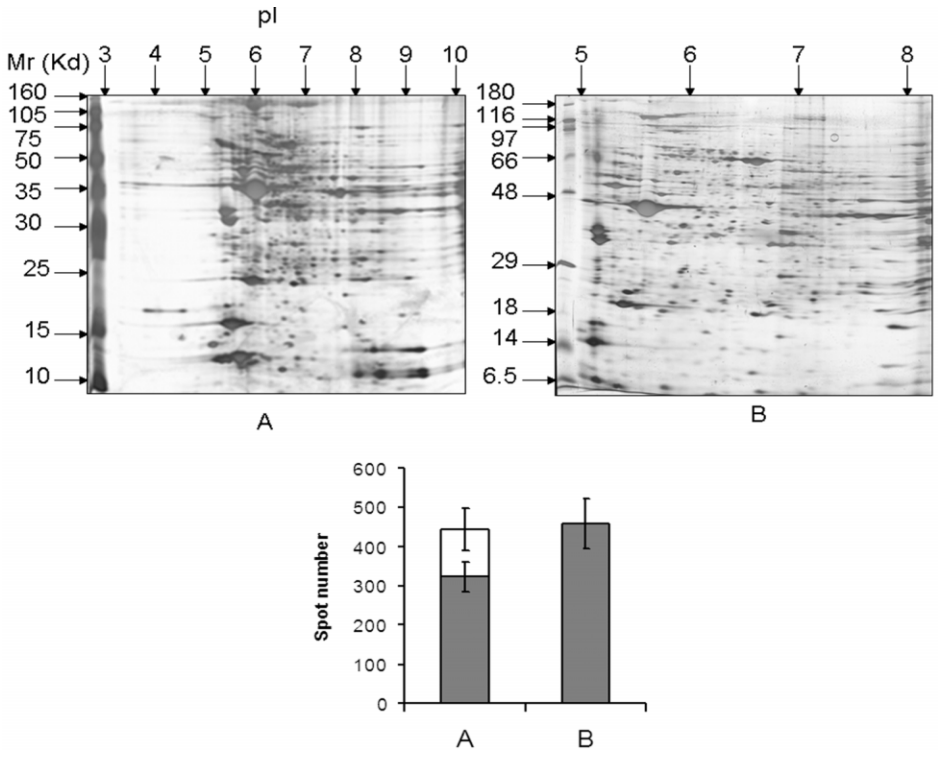


Figure 2. Distribution of protein spots from rat TA muscle using pH 3–10 broad (A) and pH 5–8 narrow (B) range IPG strips in the first dimension electrophoresis following by 12% SDS page secondary dimensional electrophoresis. Hundred microgram of total protein sample from normal control was loaded onto each gel. Gels were stained by silver. Protein molecular weight markers are present on the right side of each gel (Amersham or Sigma). Bar figure represents total spot number in pH 1–10 IPG strip gel (A) and pH 5–8 IPG strip gel (B). Dark part of bar A represents the spot number between pH 5 and 8.

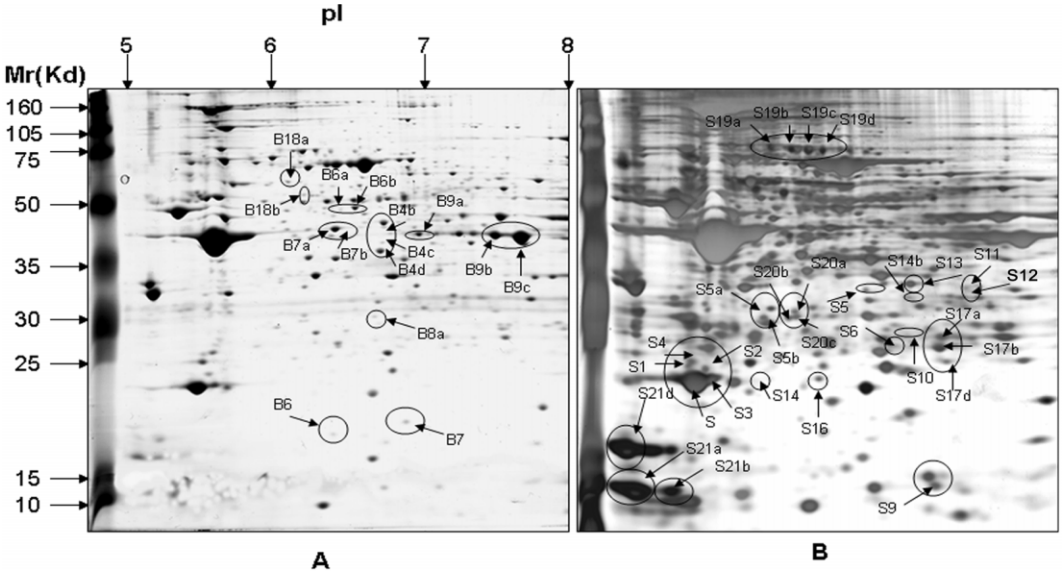


Figure 3. 2-D gel map of rat denervated TA muscle protein separated using pH 5–8 IPG strips in the first dimension electrophoresis and 12% SDS page secondary dimensional electrophoresis. Gel was stained with CBB G-250 (A) or silver (B). Protein molecular weight marker is present on right side of each gel (Amersham). Spots with significant change in densities are circled and assigned unique number. Results of protein identification are listed in Table 1.

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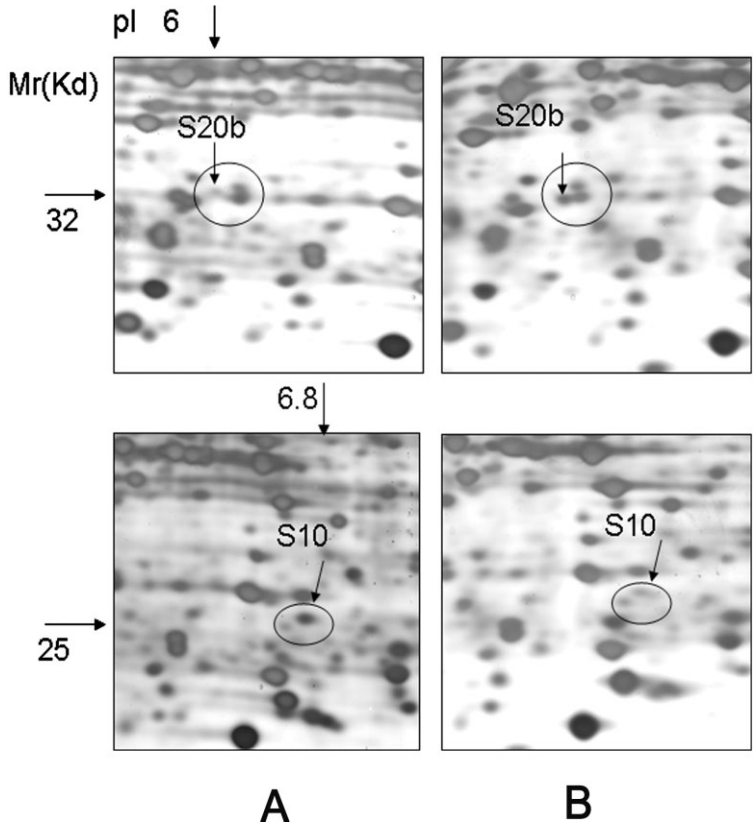


Figure 4. Representative silver stained 2-D gel images demonstrate spots with significant changes in denervated TA muscle (column B) compared with control (column A). Spot S20b is identified Proteasom 28 subunit and its density is increased by 42% in denervated TA muscle. Spot S10 is identified as OspD1 protein and its density is decreased by 41% ($p < 0.05$).

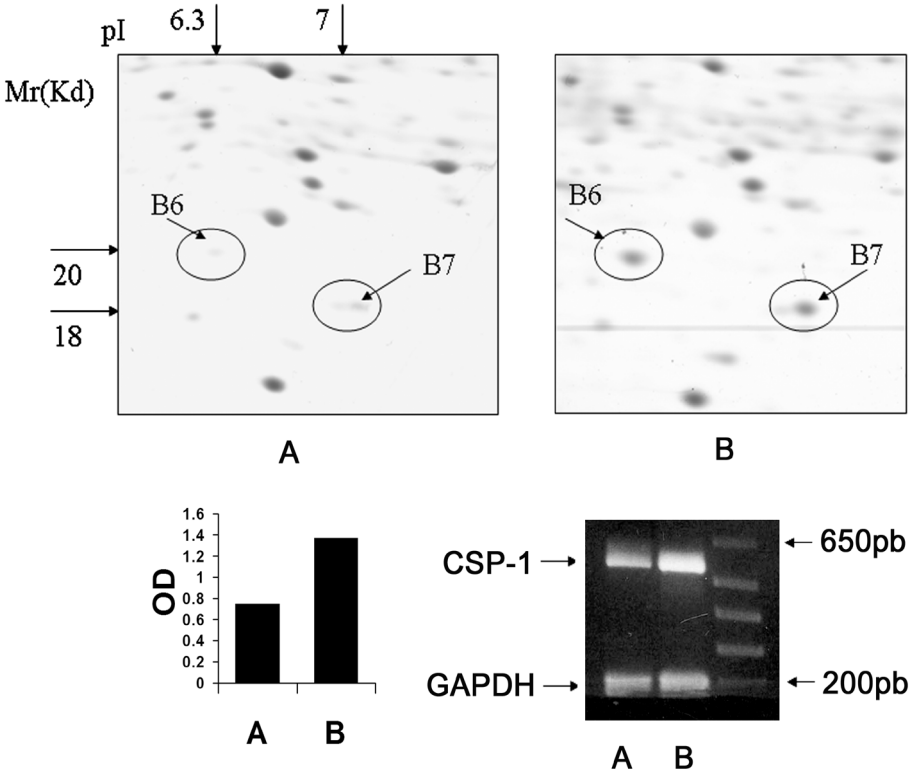


Figure 5. Alterations of expression in CSP-1 protein and its mRNA in control (A) and denervated TA muscle (B). Upper figures are CBB stained 2-D gel images of two CSP-1 protein spots (B6 and B7). Low figure is gel image of CSP-1 mRNA expression detected using RT-PCR. GAPDH mRNA is a quantitative standard control. Bar figure compares densities of mRNA bands after normalizing using GAPDH band densities.

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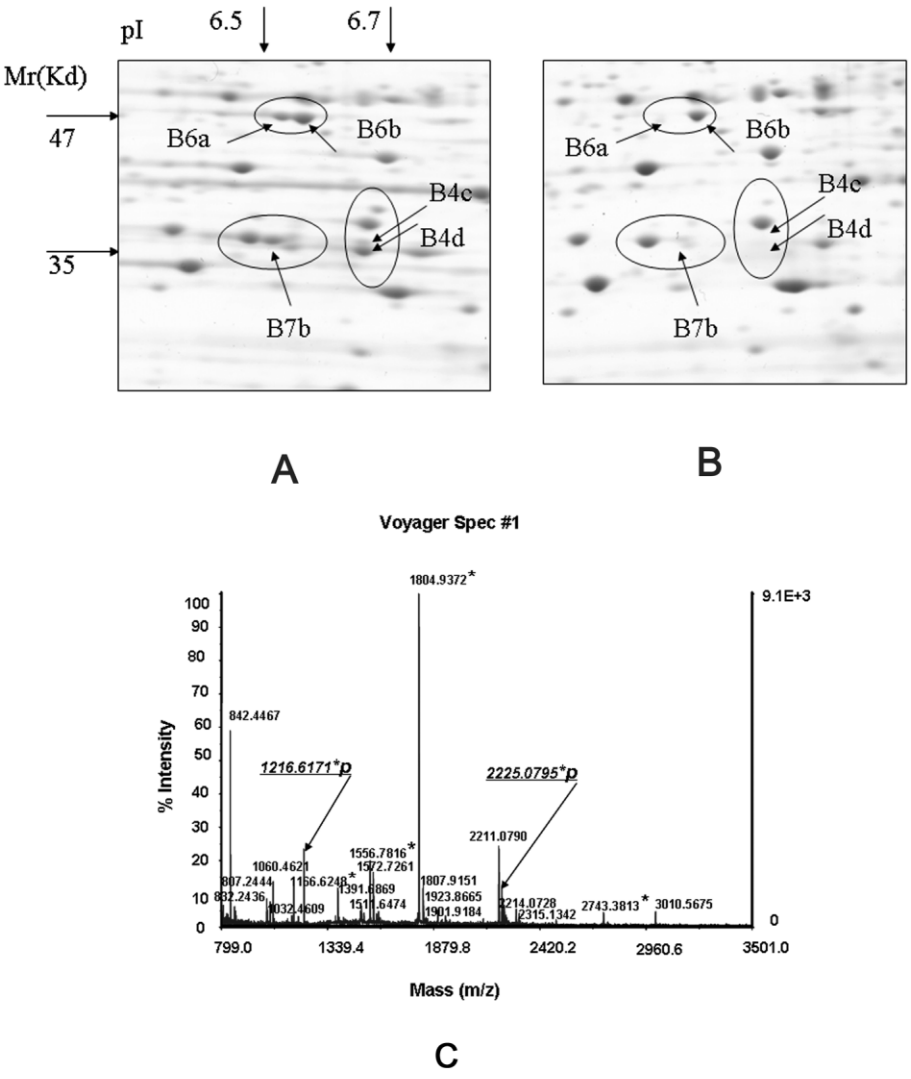


Figure 6. Multiple-spot patterns on CBB stained 2-D gels from control (A) and denervated TA muscle (B). Both horizontal spot pattern (B6a and B6b) were identified as α-enolase. Both horizontal pattern (B7b and B4c) and vertical pattern (B4c and B4d) were identified as β-enolase. Low figure is an example of spot B7b mass spectrum. All the matched mass spectrums are labeled as *. Mass spectra representing phosphorylation peptides are underlined and labeled as P. Details of phosphorylation sites of the multiple spots are listed in Table 3.

Table 2. Protein spots confirmed by LC MS/MS

Spot no.	Protein name and accession no.	Peptides matched	MOWSE scores
S	MLC 1 (gi 127131)	47	570
B6	Common salivary protein (gi 19424346)	6	153
B7	Common salivary protein (gi 19424346)	2	53
B7b	Muscle specific enolase, beta (gi 6978811)	15	495

All five β- and α-enolase spots identified in this study decreased their expression levels in denervated TA muscle. In higher vertebrates, enolase is active as a dimeric protein formed from three subunits, ubiquitous α, muscle specific β, and neuron specific γ. During ontogenesis, a unique embryonic isoform differentiates into muscle specific β isoform in striated muscle and neuron specific γ isoform in neurons, whereas the α isoform remains widely distributed in adult tissue [18, 19].

Three β-enolase heterogeneic spots present two distinct patterns on 2-D gel: a vertical pattern of two spots (B4c and B4d) similar to that observed in human muscle [18, 20] and a horizontal pattern (B7b and B4c) not previously reported. β-Enolase is a muscle tissue specific dimeric glycolytic enzyme localized to the subcellular site where ATP is produced in response to contractile demands [21]. The result of decreased β-enolase expression in this study is consistent with reports by others [22]. Differential expression between slow- and fast-twitch muscle has been observed in response to denervation, with a 50% increase in β-enolase level in slow-twitch muscle and 80% decrease in fast-twitch muscle [19, 22]. Because TA muscle possesses faster contractility

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Table 3. Phosphorylated amino acid from multiple-spot peptides

Protein name	Spot no.	Experiment MW (kDa/p*I*)	*m/z* Submitted	Detected peptide containing phosphorylated amino acid (italic and bold)
γ-tubulin complex component 2	S19a	100/6.2	1638.8224	(6)IHHDVNELL***S***LLR(18)
			2297.1580	(245)***T***FLVDPNLDL***S***IRELVHR(262)
	S19b	100/6.3	1638.8566	(6)IHHDVNELL***S***LLR(18)
			1940.9274	(77)NLDPLV***Y***LL***S***KL***T***EDK(92)[a]
			2297.1511	(245)***T***FLVDPNLDL***S***IRELVHR(262)
			1522.7662	(450)IL***ST***GKYLNVVR(461)
	S19c	100/6.4	1638.8615	(6)IHHDVNELL***S***LLR(18)
			1449.7452	(828)NF***S***AHLLDLLAR(839)
			2521.2213	(427)***Y***WDQR***YT***IVQQQIP***S***FLQK(445)[b]
	S19c	100/6.5	1638.8576	(6)IHHDVNELL***S***LLR(18)
			2297.1840	(245)***T***FLVDPNLDL***S***IRELVHR(262)
			2473.1613	(551)LEALLELALRM***ST***AN***T***DPFK(570)
β Enolase	B7b	42/6.5	1216.6171	(404)LAK***Y***NQLMR(412)
			2225.0795	(61)GVLKAVENIN***ST***LGPALLQK(80)
	B4c	42/6.7	2300.1629	(32)AAVP***S***GA***ST***GI***Y***EALELRDGDK(53)[a]
	B4d	39/6.7	1599.9353	(106)FGANAILGV***S***LAVCK(120)
α Enolase	B6a	47/6.4	1475.7394	(222)KALELLK***T***AI***S***K(233)
	B6b	47/6.6	2283.1862	(307)F***T***A***T***AGIQVVGDDL***T***VTNPKR(327)[a]
Creatine kinase	B9b	43/7.5	1675.7866	(136)***S***IKG***YT***LPPHC***S***R(148)

a) One of amino acids indicated in italic and underline is phosphorylated.
b) Two of amino acids indicated in italic and underline are phosphorylated.

than limb fast-twitch muscles, decrease of β-enolase in denervated TA may play more important role in the process of muscle atrophy.

Two α-enolase spots (B6a and B6b) were observed in a horizontal pattern similar to the pattern of expression reported in human skeletal muscle [21]. Spot B6a with an acidic p*I* has two phosphorylation sites on its peptide fragment and spot B6b with basic p*I* has one phosphorylation site. Denervation, therefore, appears to influence phosphorylation. The level of α-enolase was decreased by 94% in spot B6a and 45% in B6b. This observation is consistent with a recent report where acidic phosphorylated spots accumulated more abundantly than basic nonphosphorylated spots in regenerating muscle [23].

Five additional proteins with decreasing expression levels are Triose-phosphate isomerase (S17a), which catalyses the interconversion of dihydroxyacetone phosphate and glyceraldehyde-3-phosphate in the energy-generating glycolytic pathway [24]; NADH-ubiquinone oxidoreductase (S2), which is the first and largest complex in the electron transport chain of mitochondria [25]; isocitrate dehydrogenase (B7a), which catalyzes the conversion of isocitrate to α-oxoglutarate and carbon dioxide providing reducing equivalents for electron transport chain [26]; acetyl-coenzyme A dehydrogenase (B9a), which catalyzes the α- and β-dehydrogenation of acyl-CoAesters in fatty acid metabolism; and lactate dehydrogenase (B4b), which catalyzes formation of pyruvate from lactate produced in skeletal muscle. These changes reflect a general decline in metabolic activity and are consistent with muscle atrophy induced by immobilization where a three-fold decrease in gene expression of enzymes including ATPase, cytochrome C oxidase, and NADH dehydrogenase has been reported [16].

Both creatine kinase (B9c) and vacuolar ATP synthase subunit H (B18a) increased their expression level in denervated rat TA muscle. Creatine kinase presents two heterogeneous spots (B9b and B9c) in a horizontal pattern. The density of spot B9b (more acidic p*I*) with two phosphorylation sites was not changed significantly; however, the density of spot B9c without phosphorylation was increased by 80%. This finding is consistent with a recent study of rat slow-twitch muscle following a 4 wk course of hind limb suspension [14, 15]. ATP synthase is a commonly altered protein. Increase in its expression has been observed in denervated conditions and after hind limb suspension [15, 16].

4.2.2 Proteins associated with cell proliferation and differentiation

All five proteins identified in this category showed reduced expression levels in denervated TA muscle. Ras-like GTP binding protein (S17b) acts as a molecular binary switch that plays key roles in the control of the cellular cycle and inhibits differentiation of muscle cells [27]. 3-Hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase (S5) is a rate-limiting enzyme of the mevalonate pathway. It converts HMG-CoA to mevalonate, which plays a critical role in the prolifer-

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ation of mammalian cells [28, 29]. DNA-directed RNA polymerase alpha chain (S20c) is the core of the enzyme that catalyzes transcription of DNA into RNA, and effects skeletal muscle cell differentiation [30, 31]. DNA polymerase (B8a) is correlated with the rate of cell proliferation during perinatal development and responsible for repair type processes [32]. Replication-associated protein (S17d) is the only protein required for viral DNA replication [33]. Its role in mammalian cells has not been previously reported.

4.2.3 Protein involved in regulation of muscle contraction

Myosin light chain (MLC) plays an important role in the regulation of muscle contraction by altering the conformation of myosin heads, promoting actin-myosin interaction, and regulating myosins ATPase activity. At least seven different subunits of MLC have been observed in normal rat skeletal muscle using 2-D gels with a pH 4–7 narrow IPG strip [14]. In the current study using a pH 5–8 IPG strip, four MLC spots were separated in TA muscle, including two MLC1 spots (S and S3) and two spots identified as MLC2 (S21d) and MLC3 (S4).

Altered expression of MLC expression appears to reflect alterations in heavy chain expression [34]. Decreased expression of MLC1 has been observed in atrophied limb muscle induced by immobilization, a condition also characterized by a decrease in fast MyHC muscle [35].

In the current study, changes induced by denervation include a 42% increase in MLC1 (S) expression, and a 92% decrease in the expression of MLC1 (S3). MyHC IIA and IIX expression, frequently associated with MLC1, is increased in denervated rat TA muscle, while MyHC IIB decreases. Although not confirmed in this study, we infer that these changes are likely the result of congruent shifts in MyHC and the associated MLC.

Using pH 4–5 IPG strips, MLC2 has been separated into three horizontal spots based on differences in phosphorylation in both rabbit fast-twitch muscle and human muscle [36, 37]. In the current study, using pH 5–8 IPG strips only one MLC2 spot was identified (S21d), with a 1.5-fold increase observed in the denervated condition. Using pH 4–5 IPG strip and specific antiMLC2 antibody, three horizontal MLC2 spots from TA muscle were also detected in our recent study.

MLC3 is only expressed in fast muscle fibers [34]. Expression of MLC3 in developing limb musculature is also dependent on neural influences [38]. In the current study, MLC3 (S4) was decreased by 49% in denervated TA muscle. This observation parallels the decrease in fast IIB and EOM MyHC, and furthermore results in a decrease in the MLC3/MLC1 ratio previously shown to be associated with slower muscle contraction velocity.

Two horizontal spots of parvalbumin were increased 2.5-fold in the denervated TA muscle (S21a and S21b). Increase in its gene expression has been reported in immobilized skeletal muscle [1]. However, during aging parvalbumin expression is downregulated in rat fast-twitch skeletal muscle [39]. Parvalbumin is a high-affinity Ca^{2+} binding protein found specifically in fast-twitch muscle fibers and facilitates the rate of muscle relaxation. Gene delivery of parvalbumin causes a physiological acceleration of relaxation performance in normal hearts, and enhances relaxation performance in an animal model of slowed cardiac muscle relaxation [40]. Paryalbumin gene therapy for denervated fast type muscle, such as TA muscle, needs to be further investigated.

4.2.4 Proteins involved in the pathway of degradation of cellular proteins

The ubiquitin-proteasome pathway has been shown to be an important pathway of protein degradation in the muscle atrophy process [10, 41]. The proteasome exists as a large multisubunit protease within the cytosol (free and attach to the ER) and within the nucleus that recognizes and digests protein substrates marked for degradation by the attachment of an ubiquitin moiety. Muscle atrophy is induced in mice with over expression of the gene encoding ubiquitin ligase, whereas mice deficient of the gene are resistant to muscle atrophy [10]. In the current study, a 42% increase in the expression of proteasome 28 subunit (S20c) was observed in denervated TA muscle. This finding is consistent with the prior study reporting up-regulation of these ribosomal proteins in immobilized skeletal muscle [1].

4.2.5 Proteins associated with stress reaction

Heat shock proteins account for approximately 23% of the total protein spots identified in human TA muscle 2-D database [17]. Aging and exercise influence heat shock protein expression [42]. In the current study, expression level of heat shock transcription factor 24 (SHF24) (S20a), a specific DNA binding protein, was decreased by 55% in denervated TA muscle. This result suggests decreasing activation of stress response in denervated muscle.

4.2.6 Proteins associated with immunoreactivity and injury repair

The expression of MHC class II (S9) is increased 16-fold in denervated TA muscle, which is the largest change of expression level among the 45 protein spots identified in this study. MHC class II molecules are normally present on cell surfaces in most tissues except for the cells of the CNS and skeletal muscle [43, 44]. Muscle and neural cells, however, which do not express MHC class II under normal conditions, can be induced to express MHC class II under pathological stimulation such as inflammatory infiltrates and myasthenia gravis [45]. Expression of MHC class II has also been observed in rat skeletal muscle rendered inactive by nerve impulse blockade and denervation [46]. Nerve-evoked muscle activity, therefore, influences not only the phenotype of the muscle cells, but also their immunoreactivity.

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Purine nucleoside phosphorylase (PNP) was increased by 4.7-fold in denervated TA muscle (S14b). PNP is responsible for degradation of deoxyguanosine and related to the development of T-cell immunity [47]. Its role in skeletal muscle is not clear, however, a recent study, which found that ischemic preconditioning of myocardium was associated with activation of PNP [48], suggested that PNP may play a role in the repair process after muscle injury.

4.2.7 Proteins associated with signal transduction and membrane trafficking

Expression of C-jun-amino-terminal kinase interaction protein 1 (JIP-1) was increased two-fold in denervated TA muscle (S16). JIP-1 is a crucial component of the JNK signal pathway regulating a diverse array of biological processes including cellular survival and proliferation [49, 50]. JIP-1 limits neuronal cell death after ischemia by inhibiting phosphorylation and translocation of JNK [41, 51]. Its increase in denervated muscle may reflect changes in the regulation of cellular protection under the muscle denervation condition.

Expression of RAB-13 (S14), a smaller GTPase involved in the regulation of membrane trafficking, was decreased by 83% in denervated TA muscle. RAB-13, recruited to junctional complexes from a cytosolic pool after cell-cell contact formation [52], may have a role in regulating both the structure and function of tight junctions during TA muscle atrophy.

Other proteins associated with signal transduction undergoing change after denervation include cAMP phosphodiesterase (S18b) and Tumor-associated calcium signal transducer precursor-1 (S12). Expression of low km cAMP phosphodiesterase, which affects cellular function *via* the cAMP signaling pathway [53], increased expression threefold in the denervated condition. Tumor-associated calcium signal transducer 1 precursor, a growth factor receptor [54], was deceased by 74% in denervated TA muscle.

4.2.8 Proteins associated with microtubule nucleation

γ-Tubulin mediates microtubule nucleation on centrosomes. However, there is strong evidence that γ-tubulin has other cell functions, such as regulatory or enzymatic activity not directly related to microtubule generation [55]. In skeletal muscle, the microtubule network participates in the mechanical integration of the various organelles of the myofibers during the contraction-relaxation cycle [56]. PTM of microtubules is required for specific functions during myocyte differentiation [57]. For example, phosphorylation of γ-tubulin causes cell cycle arrest before anaphase and increases the number of cytoplasmic microtubules [55]. The pattern of four horizontal γ-tubulin spots (S19a, b, c, and d) was observed in laryngeal muscle. Increase in their expressions after denervation is associated with phosphorylation and may reflect disruption of cell cycle progression

4.2.9 Other proteins

Two spots identified as CSP-1 (B6 and B7) increased their expressions in the denervated condition. CSP-1 influences salivary gland differentiation and regulation during parotid gland development [59], however, its expression in muscle tissue has not been reported previously. Both spots of CSP-1 in TA muscle were dissimilar to those identified in salivary gland with respect to MW and p*I* [58]. Further confirmation has been performed in the current study using LC MS/MS and RT-PCR in the same specimens. Although these results suggested that TA muscle tissue expresses CSP-1, contamination of CSP 1 from adjacent minor salivary gland cannot be discounted.

The level of 5,10 methylterahydrofolate reductase (S5b), a key enzyme in homocysteine metabolism, decreased by 48% in denervated TA muscle. Its role in the process of muscle atrophy induced by denervation is not known.

Other proteins, changing expression levels in the denervated TA muscle were identified as OspD1 (S10), Pir protein (S5a), membrane protein 7 (S1), and unknown protein spots S11, S6, and S13. Function of these proteins need to be further studied.

4.3 Conclusion

The application of proteomics has provided new opportunity to assess the molecular nature of muscle including expression level, PTM, protein–protein interaction and subcellular location. This study represents the first application of proteomics for the evaluation of laryngeal muscle applied to a pathological condition. Change in cellular protein expression is anticipated to occur in the denervated condition and has been demonstrated previously with MyHC components. Global cellular protein changes may represent a more sensitive marker for the status of laryngeal denervation. Recognizing that muscle mass is decreased in the denervated condition, normalization of protein concentration and application of equal amounts of total protein may yield relative increases in the remaining pool of proteins if one or more abundant proteins present in the normal condition is lost. Conclusions regarding upregulation are therefore tempered by this potential concentration effect. Regardless of having established this initial database, our intent is to monitor proteomic shifts in response to therapeutic gene transfer. Further development of this database will provide a valuable resource for the molecular analysis of normal and pathologic conditions affecting human striated muscle, and create a window for assessing treatment responses based on changes in the muscle cell proteome.

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